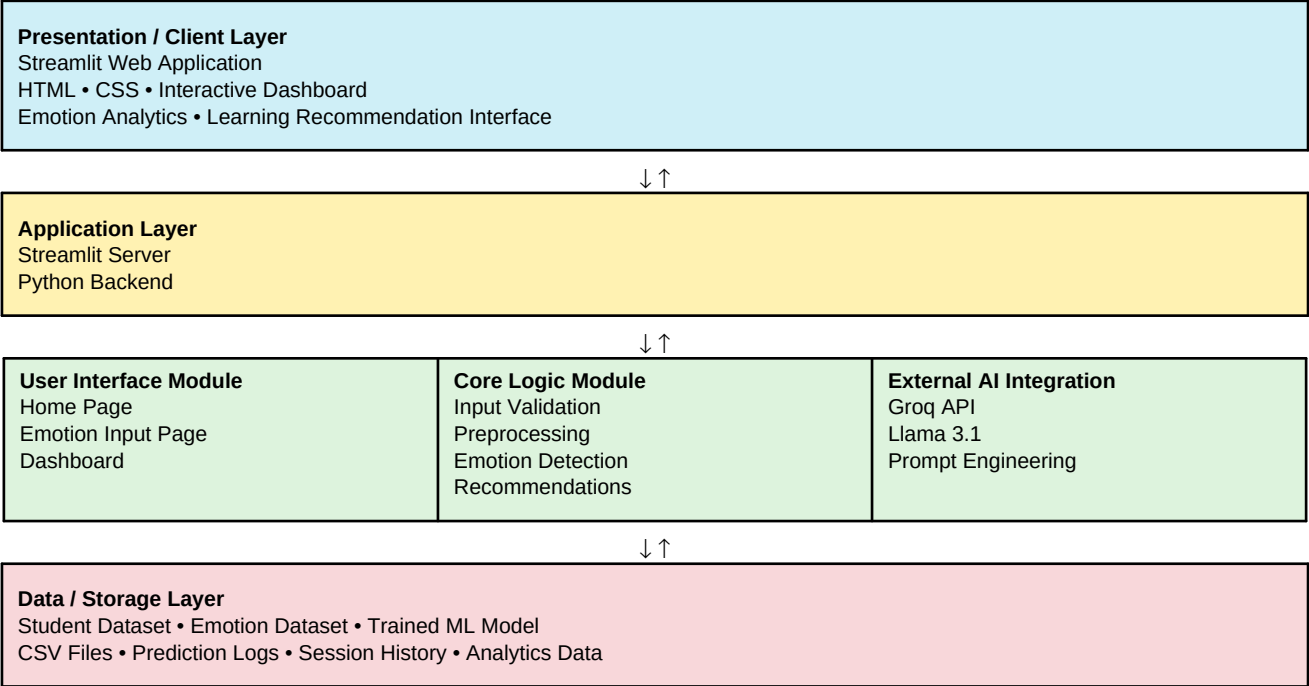


Solution Architecture

Date	
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks



Component Description Table

Component	Description	Technologies Used
Presentation Layer	Interactive Streamlit UI	Streamlit, HTML, CSS
Backend Layer	Routing, preprocessing and business logic	Python
Emotion Detection Engine	TF-IDF + Naive Bayes emotion detection	Scikit-learn, NLP
Generative AI Module	Personalized guidance	Groq API (Llama 3.1)
Analytics Module	Emotion trends and dashboards	Pandas, Plotly
Data Storage	Datasets, logs and history	CSV, Pandas